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Track 2 — Advanced Machine Learning Internship

Alignment With Reality Is Not Optional: A Channel-Normalized Critique of Deep Sequential Imputation on PhysioNet 2012

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Abstract

Published benchmarks report state-of-the-art (SOTA) point-imputation MAE near 0.24 for deep sequential imputers (BRITS, SAITS) on PhysioNet 2012. This figure is a micro-average pooled across all masked points regardless of channel; because dense vitals (heart rate, blood pressure) constitute the large majority of observed points in this dataset, such a metric is structurally weighted toward the easiest channels to reconstruct. We re-evaluate BRITS and SAITS under a channel-normalized, macro-averaged MAE that weights all 36 clinical channels equally, on an identical held-out cohort of 800 patients (53,556 masked points), under a strict patient-level split, validation-based early stopping, gradient-clipped optimization, and physiologically-bounded output clamping. Under this metric, BRITS (0.7456) and SAITS (0.7521) both underperform a trivial constant-mean-imputation baseline (0.6687), while deterministic baselines — Linear Interpolation (0.3849), LOCF (0.4312), k-NN (0.4128), MICE (0.4284) — lead by a wide margin on both pointwise and distributional fidelity. Per-channel diagnostics reveal two coexisting, distinguishable error patterns: genuine predicted-variance collapse on a subset of channels, and a separate metric-sensitivity effect tied to channels with implausibly large ground-truth variance, most notably arterial pH, whose reported standard deviation ($\approx$23 pH units in two independent runs) exceeds physiological possibility and is most plausibly explained by uncorrected data-entry artifacts native to the PhysioNet 2012 release. We extend the evaluation with a stratified error analysis by mortality outcome, finding that all seven methods, without exception, reconstruct non-survivors' physiology less accurately than survivors' — a consistent, method-independent disparity with direct relevance to equitable model deployment.

A second extension evaluates downstream classification utility (AUROC/AUPRC) of a mortality predictor trained on each method's reconstructed data. Unlike the primary benchmark, where deterministic methods lead decisively and both deep sequential imputers underperform even a trivial mean-imputation baseline, the downstream-utility results show no consistent method-class advantage: overall predictive power for in-hospital mortality remains modest across all seven methods, and neither AUROC nor AUPRC rankings track each method's pointwise reconstruction accuracy. We treat this as further evidence, alongside the primary benchmark, that pointwise, channel-weighted, and task-level evaluation criteria can yield materially different — and at times uncorrelated — conclusions about the same set of methods, consistent with our broader argument that rigor in evaluation, not the sophistication of the model under evaluation, is the operative bottleneck in this literature.

Keywords: time-series imputation, ICU physiological data, PhysioNet 2012, evaluation metric bias, BRITS, SAITS, reproducibility, clinical machine learning

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1 Introduction

1.1 Motivation and Problem

Continuous physiological monitoring in Intensive Care Units (ICUs) frequently experiences data loss due to sensor detachment, patient movement, routine clinical interventions, and telemetry latency. Missingness in this setting is heterogeneous: it spans protocol-driven gaps (scheduled labs), clinically-driven gaps (sensors removed during acute intervention), and informative sequence truncation (discharge, transfer, or death). Deep sequential imputation models — BRITS [1] and SAITS [2] chief among them — report substantial accuracy gains over classical statistical baselines on PhysioNet 2012 [3], a widely used 48-hour multivariate ICU benchmark. These reported gains are computed as a single pooled (micro-averaged) error statistic across all masked data points. Because PhysioNet 2012's 36 clinical channels differ by roughly an order of magnitude in how densely they are sampled — continuous vitals versus labs drawn once or twice per 48-hour window — a micro-averaged statistic is mathematically dominated by whichever channels contribute the most masked points, which in this dataset are precisely the easiest channels to reconstruct.

1.2 Research Question

This work asks a narrow, falsifiable question: does the reported advantage of deep sequential imputers over classical baselines persist under an evaluation metric that weights all clinical channels equally, rather than weighting by how often each channel is observed? A secondary, methodological question emerged during execution and is treated with equal seriousness in this paper: how much of an imputation benchmark's apparent result is attributable to the evaluation pipeline itself, rather than to the models under comparison?

1.3 Scope and Contributions

1.3.1 Scope

The evaluation cohort is PhysioNet 2012 Set-A (approximately 4,000 patients), split patient-wise into training, validation, and test folds (60/20/20). Two deep sequential imputers (BRITS, SAITS) are compared against five classical baselines (Mean Imputation, LOCF, Linear Interpolation, k-NN, MICE) under a 20% random-point synthetic masking protocol.

1.3.2 Contributions

1. A channel-normalized, macro-averaged MAE metric, applied identically across seven imputation methods on an identical held-out cohort, demonstrating that the reported literature advantage of deep sequential imputers does not reproduce under this metric.
2. A validation-based early-stopping procedure for the neural imputers, verified against training logs to be genuinely driven by validation loss rather than training loss.
3. A seed-controlled experimental design (verified to produce bit-identical model outputs under a fixed seed) intended to isolate the effect of gradient clipping from ordinary training-run variance.
4. Two clinically motivated evaluation extensions — stratified error by mortality outcome, and downstream classification utility — yielding a clean, method-independent stratified finding and a null result on method-class separation in downstream utility.
5. An explicit, itemized reconciliation between the evaluation scope originally proposed for this project and what has been executed to date.

1.3.3 Explicitly Out of Scope

This work does not implement or evaluate: Mahalanobis-distance-based multivariate physiological plausibility checking (only univariate per-channel clamping is implemented); the TIMBA/Mamba architecture family; RMSE, MRE, or Wasserstein distance as evaluation metrics; Cubic Spline interpolation as a classical baseline; or evaluation on the full 12,000-patient PhysioNet 2012 release (Sets A, B, and C combined). Each omission is stated explicitly in Section 9 rather than left for a reader to infer.

1.4 Report Structure

Section 2 surveys the relevant literature on time-series imputation and evaluation methodology. Section 3 formalizes the imputation task. Section 4 describes the dataset and associated ethical considerations. Section 5 details the full experimental methodology. Section 6 reports results, including downstream-utility results. Section 7 analyzes error patterns in detail, including a data-quality finding independently reproduced across two separate experimental runs. Section 8 discusses the implications. Section 9 states limitations explicitly. Section 10 proposes future work, and Section 11 concludes.

2 Related Work

2.1 Deep Learning for Time-Series Imputation

BRITS [1] models multivariate time series as a bidirectional recurrent dynamical system, using learned time-decay parameters to treat the interval since a channel's last observation as an explicit input feature, allowing gaps to be modeled as informative rather than incidental. SAITS [2] instead uses diagonally-masked multi-head self-attention, jointly modeling temporal position and cross-channel correlation without recurrence, at the cost of quadratic memory scaling in sequence length. Both are implemented and benchmarked in this work via the open-source PyPOTS toolkit [4].

2.2 Classical and Statistical Imputation Baselines

Last Observation Carried Forward (LOCF) and linear interpolation remain widely used in clinical time-series pipelines for their simplicity and their implicit anchoring to a specific patient's own observed trajectory. k-Nearest Neighbors imputation and Multiple Imputation by Chained Equations (MICE) instead exploit cross-sectional feature correlation but have no native mechanism for sequential temporal dependency, an acknowledged limitation in the literature that motivates the sequential deep learning approaches above.

2.3 Evaluation Metrics for Imputation Quality

The imputation literature has historically reported Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) as pooled statistics over all masked points in a test set. This convention is rarely examined critically in the context of highly heterogeneous multivariate clinical data, where channels differ by orders of magnitude in observation frequency. The present work's central methodological contribution is to make this pooling choice an explicit experimental variable rather than an implicit convention.

2.4 PhysioNet 2012 as a Benchmark

The PhysioNet/Computing in Cardiology Challenge 2012 dataset [3] provides 48-hour multivariate ICU records across 36 time-varying clinical variables and 5 static descriptors, originally released for in-hospital

mortality prediction. Its widespread reuse as an imputation benchmark is a secondary use of a dataset designed for a different primary task, a point with direct bearing on this paper's downstream-utility evaluation tier (Section 5.9.2).

2.5 Gap Addressed by This Work

No evaluation of BRITS or SAITS on PhysioNet 2012 known to the authors reports a channel-equalized error metric, a validation-based early-stopping correctness check, or a seed-controlled ablation of training stabilization techniques. This work addresses all three, and in doing so surfaces methodological considerations that are, in the authors' view, at least as significant a finding as the headline benchmark comparison itself.

3 Problem Formulation

3.1 Task Definition

The task is multivariate time-series imputation, formulated as self-supervised sequence reconstruction. Given an observed matrix $X \in \mathbb{R}^{T \times D}$ ($T = 48$ hourly steps, $D = 36$ channels), a binary observation mask $M \in \{0, 1\}^{T \times D}$, and a time-lag matrix $\Delta \in \mathbb{R}^{T \times D}$ recording elapsed time since each channel's last valid observation, the objective is to produce $\hat{X} \in \mathbb{R}^{T \times D}$ such that $\hat{X} \odot M = X \odot M$ (known values are preserved exactly) while unobserved entries are filled with plausible estimates.

3.2 Input/Output Spaces

The unit of prediction is a single reconstructed physiological state vector $\hat{x}_t \in \mathbb{R}^D$ at discrete hour t. Informative sequence-truncation (right-censoring due to discharge, transfer, or death) is formally excluded from this within-sequence reconstruction formulation, as it requires joint survival-analysis machinery outside this project's scope.

3.3 Missingness Taxonomy

Missingness in ICU telemetry is heterogeneous rather than uniform: protocol-driven missingness (e.g., labs drawn once per shift) behaves approximately as Missing At Random conditional on time; clinically-driven missingness (e.g., sensor removal during acute intervention) behaves as Missing Not At Random; and sequence truncation is informative and excluded from scope as above. This taxonomy motivates evaluating reconstruction fidelity across the full heterogeneity of channel types rather than on a single representative subset.

4 Dataset and Ethical Considerations

4.1 Dataset Overview

PhysioNet 2012 Set-A comprises 4,000 adult ICU admissions, each with up to 48 hours of timestamped vitals and laboratory measurements across 36 variables, plus five static descriptors (Age, Gender, Height, ICUType, Weight) excluded from the time-series tensor. An accompanying outcomes file provides an `In-hospital_death` binary label per patient, used exclusively in the Section 6.4–5.9.2 clinical evaluation extensions and never exposed to any imputation model during training.

4.2 Preprocessing Pipeline

Raw per-patient records are parsed, pivoted to a Parameter $\times$ Time matrix, and discretized into fixed 48-hour, 1-hour-resolution grids via mean aggregation within each hour. Patients are split at the patient level (never at the timestep level) into training (60%), validation (20%), and test (20%) folds prior to any statistic computation, masking, or scaling, eliminating a specific and well-documented class of data leakage in time-series benchmarking.

4.3 Licensing

PhysioNet 2012 is released under a data use agreement permitting research use; this work does not redistribute any patient-level data, and all reported figures are aggregate statistics computed over the held-out test cohort.

4.4 Personal and Sensitive Information

All records are drawn from a de-identified public research release. No re-identification attempt of any kind was made or is possible from the data or artifacts produced by this project.

4.5 Bias and Representation

PhysioNet 2012 Set-A's demographic composition is not independently re-audited in this work; the stratified evaluation in Section 6.4 examines disparity across mortality outcome specifically, not across demographic subgroup, and this distinction is treated as an explicit limitation (Section 9).

4.6 Summary

The dataset is used strictly for its designed research purpose (mortality-linked physiological time series), under patient-level isolation throughout, with no attempt to extend its use beyond aggregate, de-identified analysis.

5 Methodology

5.1 Pipeline Overview

Patient-level partitioning precedes all downstream statistic computation, masking, and model fitting, with the validation fold used exclusively for early-stopping model selection and the test fold used exclusively for final reporting: Raw PhysioNet 2012 Set-A $\rightarrow$ Patient-Level Train (60%) / Validation (20%) / Test (20%) Split $\rightarrow$ [Train fold: statistics, model fitting (natural missingness only)] / [Validation fold: synthetic masking; early stopping only] / [Test fold: synthetic masking; sole source of all reported metrics].

5.2 Data Splitting Strategy

An 800-patient test fold (20% of Set-A) is held out and used exclusively for final reporting; a separate 20%-of-remainder validation fold is used exclusively for early-stopping checkpoint selection and is never used to compute any number reported in Section 6.

5.3 Synthetic Masking Protocol

A 20% random-point synthetic mask is applied independently to the validation and test folds: for each patient, 20% of all currently-observed cells (pooled across channels) are randomly selected, their true values recorded as ground truth, and the cells set to missing. The training fold receives no synthetic mask and is fit on natural missingness only.

5.4 Heavy-Tailed Channel Transform

Four channels with strongly right-skewed, heavy-tailed raw-unit distributions (AST, ALT, Bilirubin, Urine) receive a $\log 1p$ transform prior to z-scoring, intended to stabilize gradient magnitudes during neural network training. Normalization statistics (μ, σ per channel) are computed once on the transformed training fold and applied identically to validation and test data.

5.5 Physiological Bounds and Output Clamping

All model outputs, deep learning and classical alike, are clamped to a fixed per-channel physiological range (Table 7 in Appendix) prior to any metric computation, as an engineering guardrail against implausible extrapolation. These bounds are wide safety margins, not clinical reference ranges, and have not been reviewed by a clinician.

5.6 Classical Baselines

Five classical methods are implemented: Mean Imputation (global training-fold mean); LOCF (forward-fill, with backward-fill and mean-fill fallback for leading gaps); Linear Interpolation (with the same fallback chain); k-Nearest Neighbors ($k = 5$, operating in the same $\log 1p$-transformed, z-scored space as the neural models for a fair comparison); and MICE (Iterative Imputer, 10 iterations, same transformed space as k-NN).

5.7 Deep Learning Models

5.7.1 Early-Stopping Procedure

Neural imputers are trained with validation data passed explicitly as `val_set={"X": ..., "X_ori": ...}`, the format PyPOTS requires internally to compute a genuine validation loss, so that early-stopping patience and best-checkpoint selection are driven by validation loss rather than training loss. Training logs confirm this: BRITS terminates via patience at epoch 40 with its best checkpoint restored from epoch 30; SAITS terminates at epoch 49 with its best checkpoint from epoch 39 — in both cases, an earlier epoch than the point of termination, consistent with genuine validation-loss-driven model selection.

5.7.2 Gradient Clipping

A custom `ClippedAdam` optimizer (gradient-norm clipping, max_norm $= 1.0$) was implemented after an early attempt to combine the $\log 1p$ transform with an elevated learning rate exhibited unstable gradients. `ClippedAdam` subclasses Adam and overrides `.step()` to clip before delegating to the parent implementation, and was verified to train successfully end-to-end via `BRITS.fit()`/`SAITS.fit()` prior to use in this pipeline.

5.7.3 Seed Control and Reproducibility

`pypots.utils.random.set_random_seed` was independently verified, via a controlled two-run comparison, to produce bit-identical model outputs at a fixed seed and measurably different outputs un-

der a different seed. All models reported in this paper are trained with a single fixed seed (42); a designed, code-verified ablation comparing `ClippedAdam` against plain Adam under this seeding scheme exists in the project pipeline but was not executed in the run this paper reports (Section 5.10).

5.8 Evaluation Metrics

5.8.1 Channel-Normalized Macro-Averaged MAE

For each of the 36 channels, MAE is computed over that channel's masked test points and divided by that channel's standard deviation (computed over the same evaluation-set points, unless otherwise noted); the 36 resulting values are averaged with equal weight regardless of how many points contributed to each. This directly targets a specific property of PyPOTS' own training loss and validation metric: both are implemented as $\mathrm{sum}(\mathrm{error} \times \mathrm{mask})/\mathrm{sum}(\mathrm{mask})$, a single pooled statistic with no channel weighting.

5.8.2 KS-Statistic

The Kolmogorov–Smirnov statistic between the pooled distribution of all imputed values and the pooled distribution of all true values is reported as a distributional-fidelity measure. This statistic, as currently computed, pools all 36 channels into a single comparison and likely carries a volume-weighting bias analogous to the one corrected for MAE; this is stated as an explicit limitation (Section 9), not resolved in this paper.

5.9 Extended Clinical Evaluation Tiers

5.9.1 Stratified Error by Mortality Outcome

For each method, channel-normalized macro-averaged MAE is computed separately for patients who survived their ICU stay and those who did not, using each channel's full-test-set standard deviation (not a stratum-specific one) as the shared denominator for both strata. This choice is deliberate: splitting into strata roughly halves the sample count per channel, and several channels already have small overall n in the full test set; a stratum-specific denominator for such a channel would be estimated on a handful of points and would add noise rather than signal. Holding the denominator fixed means any observed difference between strata reflects a genuine difference in the numerator (reconstruction error), not a moving, noisier denominator.

5.9.2 Downstream Clinical Utility

For each method, a compact per-patient feature vector (the mean and standard deviation of each of the 36 channels across the 48-hour window) is extracted from the fully reconstructed tensor, and a Random Forest classifier is trained under stratified 5-fold cross-validation to predict in-hospital mortality, with class weighting applied for the observed 13.4% outcome imbalance. One shared fold partition is held fixed across all seven methods so that differences in reported AUROC/AUPRC are attributable to reconstruction quality rather than to which patients fall into which fold.

5.10 Ablation Execution Status

A gradient-clipping ablation (`ClippedAdam` vs. plain Adam, same seed, same architecture hyperparameters) was designed and its supporting code independently verified in earlier development of this pipeline. It was not executed in the run this paper reports, and no ablation numbers from a matching hyperparameter configuration are available; reusing previously-obtained ablation figures from a different hyperparameter

configuration (smaller hidden-layer width) would misrepresent them as applying to the model configuration reported here, and this paper does not do so.

5.11 Design Decisions Summary

Table 1 summarizes the principal design decisions and their rationale.

Decision	Rationale
Patient-level split before any statistic/mask	Eliminates leakage from statistics or masking computed across fold boundaries
Channel-normalized macro MAE	Prevents high-volume channels from dominating the pooled statistic
Explicit `val_set` construction	Ensures early stopping is genuinely driven by validation loss (Sec. 5.7.1)
Gradient clipping	Addresses observed training instability under the $\log 1p$ transform
Fixed-seed training	Enables attribution of any observed effect to a deliberate change, not initialization noise
Shared CV fold partition across methods	Ensures downstream-utility differences reflect reconstruction quality, not fold composition

Table 1: Summary of principal methodological decisions.

6 Results

6.1 Experimental Setup

All results below are computed on the identical 800-patient held-out test fold (53,556 masked points across 36 channels), for all seven methods, under the metrics and clamping defined in Section 5. BRITS was trained for 40 epochs (patience-stopped; best checkpoint epoch 30, RNN hidden size 128, 236,192 trainable parameters). SAITS was trained for 49 epochs (patience-stopped; best checkpoint epoch 39, $d_{\text{model}} = 256$, 1,639,280 trainable parameters).

6.2 Primary Benchmark

Imputation Strategy	Macro-Norm MAE (σ_{eval})	Macro-Norm MAE (σ_{train})	KS-Statistic
Linear Interpolation	0.3849	9.6778	0.0108
k-NN Imputation ($k = 5$)	0.4128	12.9750	0.0182
MICE (Iterative)	0.4284	12.4830	0.0174
LOCF (Forward-Fill)	0.4312	11.1956	0.0050
Mean Imputation	0.6687	21.6537	0.1318
BRITS (RNN)	0.7456	21.3209	0.0446
SAITS (Attention)	0.7521	21.4922	0.0510

Table 2: Primary benchmark, identical 800-patient held-out test cohort (53,556 points, 36 channels), all seven methods. Lower is better throughout.

Both deep sequential imputers underperform the trivial Mean Imputation baseline on the primary channel-normalized metric, and underperform every deterministic baseline by a substantial margin. Deterministic

methods also lead decisively on distributional fidelity (KS-statistic), with LOCF and Linear Interpolation an order of magnitude closer to the true pooled distribution than BRITS or SAITS.

6.3 Per-Channel Diagnostic Audit

The two architectures share four of five channels in each category (inflation and collapse), a pattern examined further in Section 7.

Channel (BRITS)	**MAE**	**true σ**	**pred σ**	**var. ratio**
Top 5 channels driving macro-MAE inflation				
NISysABP	24.88	22.26	22.19	0.997
MAP	16.71	15.72	14.94	0.950
Na	5.51	5.26	4.66	0.886
WBC	6.39	6.52	4.76	0.729
HCT	5.12	5.26	4.29	0.815
Top 5 channels by variance collapse (lowest variance ratio)				
SysABP	18.52	24.13	0.190	0.0079
TroponinI	6.92	10.38	0.246	0.0237
AST	616.98	2262.39	58.41	0.0258
pH	1.45	23.79	0.615	0.0259
Temp	0.78	1.98	0.386	0.1946

Table 3: BRITS per-channel diagnostic audit (held-out test fold).

Channel (SAITS)	**MAE**	**true σ**	**pred σ**	**var. ratio**
Top 5 channels driving macro-MAE inflation				
MAP	17.82	15.72	16.69	1.062
NISysABP	25.14	22.26	22.68	1.019
Na	5.59	5.26	4.73	0.898
HCT	5.17	5.26	4.57	0.868
WBC	6.36	6.52	4.51	0.691
Top 5 channels by variance collapse (lowest variance ratio)				
SysABP	18.50	24.13	0.093	0.0039
AST	618.19	2262.39	45.77	0.0202
pH	1.48	23.79	0.638	0.0268
TroponinI	6.82	10.38	0.306	0.0295
Temp	0.80	1.98	0.416	0.2097

Table 4: SAITS per-channel diagnostic audit (held-out test fold).

6.4 Stratified Error by Mortality Outcome

The test cohort's mortality rate is 13.4% (107 of 800 patients).

Every one of the seven methods, without exception, reconstructs non-survivors' physiology less accurately than survivors'. The magnitude of this gap varies substantially by method — Mean Imputation shows by far the largest disparity, consistent with non-survivors' physiology being more erratic and thus worse-served by a single constant — but the direction is universal.

Method	Survivors	Non-Survivors	Δ (Non-Surv. $-$ Surv.)
Linear Interpolation	0.3790	0.4132	+0.0342
k-NN ($k = 5$)	0.4023	0.4614	+0.0591
MICE	0.4186	0.4763	+0.0577
LOCF	0.4246	0.4640	+0.0394
Mean Imputation	0.6372	0.8166	+0.1794
BRITS	0.7446	0.7550	+0.0104
SAITS	0.7507	0.7664	+0.0157

Table 5: Channel-normalized macro MAE stratified by mortality outcome, shared full-test-set σ denominator (Sec. 5.9.1).

6.5 Downstream Clinical Utility

Table 6 reports AUROC and AUPRC for a mortality classifier trained on each method's reconstructed data, following the protocol of Section 5.9.2.

Method	AUROC (mean $\pm$ std)	AUPRC (mean $\pm$ std)
BRITS	0.8013 ± 0.0455	0.3570 ± 0.0487
Linear Interpolation	0.7798 ± 0.0496	0.3481 ± 0.0453
SAITS	0.7767 ± 0.0395	0.3257 ± 0.0590
k-NN ($k = 5$)	0.7742 ± 0.0534	0.3397 ± 0.0535
LOCF	0.7676 ± 0.0633	0.3525 ± 0.0655
Mean Imputation	0.7630 ± 0.0529	0.3793 ± 0.1062
MICE	0.7555 ± 0.0484	0.3312 ± 0.0339

Table 6: Downstream mortality-classification utility. Higher is better throughout.

Unlike the primary benchmark (Section 6.2), where deterministic methods led decisively and both deep sequential imputers underperformed even the trivial Mean Imputation baseline, the downstream-utility results do not reproduce that separation. All seven methods fall within a comparatively narrow AUROC band (0.7555–0.8013) with substantially overlapping standard deviations (0.0395–0.0633), and the ranking by AUPRC does not track the ranking by AUROC or by pointwise MAE: Mean Imputation, the weakest method by a wide margin on Section 6.2's channel-normalized MAE, attains the highest AUPRC (0.3793) of any method, while BRITS attains the highest AUROC (0.8013) despite ranking among the two weakest methods on pointwise MAE.

Given the overlap between methods' standard deviations, we do not read Table 6 as evidence that any one method or method class is better suited to downstream classification than another; the differences observed are more plausibly attributable to fold-to-fold variance in the cross-validation splits than to a genuine, method-dependent effect. Two observations follow instead. First, the compact per-patient summary used for classification (the mean and standard deviation of each of the 36 channels over the 48-hour window, Section 5.9.2) evidently retains enough signal for mortality prediction regardless of which of the seven imputation strategies produced the underlying reconstruction. Second, a method's advantage on pointwise reconstruction fidelity does not transfer in any consistent way to downstream classification utility in this evaluation — consistent with the point made in Section 8.1 that pointwise, channel-weighted, and task-level metrics can answer different empirical questions, and a result favorable under one need not be favorable, or even distinguishable, under another.

7 Error Analysis

7.1 Variance Collapse

Both architectures show a cluster of channels (SysABP, TroponinI, AST) with predicted standard deviation below 3% of true standard deviation — consistent with conditional-mean-seeking behavior under an elementwise training objective when a channel's available training signal is thin. SysABP's collapse is the most extreme observed in either model (variance ratio 0.0079 for BRITS, 0.0039 for SAITS), notably more severe here than in an earlier iteration of this pipeline where PaCO2, Cholesterol, and ALP occupied the equivalent position — indicating that which specific channels collapse is sensitive to model configuration (this run uses a wider BRITS hidden layer and wider SAITS model dimension than the earlier iteration), even though the collapse phenomenon itself recurs across configurations.

7.2 A Reproduced Data-Quality Finding: Arterial pH

Arterial blood pH is one of the most tightly homeostatically-regulated variables in human physiology; a population standard deviation on the order of 0.05–0.3 pH units is physiologically expected. This run reports a test-set pH standard deviation of 23.79, and an independent earlier run of a related pipeline configuration reported 22.93 — two separate experimental runs, on overlapping but not identical model configurations, independently producing a standard deviation for this channel roughly two orders of magnitude larger than physiological plausibility permits. This is most parsimoniously explained by uncorrected data-entry artifacts in the PhysioNet 2012 raw release rather than by any property of the models under evaluation, and constitutes the strongest evidence in this project for a ground-truth data-quality concern that Section 9 identifies as unresolved.

The practical consequence differs by run in an instructive way. In an earlier run, pH's inflated denominator combined with genuinely large model error to produce a normalized MAE near 10 (an apparent inflation finding). In the present run, pH instead appears among the lowest-error collapse channels (normalized MAE 0.06), because the same implausibly large denominator this time deflates an error that would otherwise appear larger. The same underlying data artifact thus distorts the same channel's normalized score in opposite directions depending on incidental interaction with model error magnitude — a concrete illustration that a channel-normalized metric is only as trustworthy as the ground truth it normalizes against.

7.3 Root Causes

Combining Sections 7.1–7.2 with the primary benchmark, three distinguishable, non-mutually-exclusive root causes are implicated: (i) uneven per-channel training signal under a pooled, unweighted training loss; (ii) nonlinear error amplification specific to the four $\log 1p$-transformed heavy-tailed channels, where a small transformed-space error can correspond to a large raw-unit error for large true values; and (iii) ground-truth data-quality artifacts independent of any model, evidenced concretely by the pH finding above. No single cause explains the full pattern; all three appear to contribute simultaneously across different subsets of channels.

8 Discussion

8.1 The Micro/Macro Averaging Critique

Under a metric that weights all 36 channels equally rather than weighting by observation volume, the literature-reported advantage of deep sequential imputers over classical baselines does not reproduce in this

evaluation. This is best characterized as evidence that pooled, point-weighted MAE and channel-weighted MAE answer different empirical questions — not as evidence that prior reported figures are computed incorrectly on their own terms.

8.2 Why Deterministic Methods Lead Under Random-Point Masking

A 20% random-point mask predominantly produces short, often single-timestep gaps. Linear interpolation and LOCF are anchored to a specific patient's own adjacent real observations and default to a population mean only in the small number of genuinely underdetermined cases, giving these methods a bounded worst case that cannot fall far below constant-mean performance. Sequential neural imputers trained via a pooled elementwise loss have no analogous floor, and both BRITS and SAITS fall below it in this evaluation.

8.3 Refuted and Unresolved Hypotheses, Reported Honestly

Earlier development of this pipeline tested, and did not confirm, the hypothesis that gradient clipping induces variance collapse by suppressing confident extreme predictions; a seed-controlled paired comparison found the opposite direction of effect for the two channels most associated with collapse. That finding is not reproduced with the present model configuration (Section 5.10) and is reported here as an open question rather than a settled one. Similarly, the absence of a consistent method-class separation in downstream-utility results (Section 6.5) is reported as an open finding rather than smoothed into the paper's headline narrative: it is not explained by any hypothesis this paper is able to confirm, and is left as such.

9 Limitations

1. The optimizer ablation was not executed against the model configuration reported in this paper (Section 5.10); a prior finding under a different hyperparameter configuration is not reused here to avoid misattributing it to the present results.
2. Single-seed training throughout. All primary-benchmark and Task 1 results reflect one fixed seed per model. The seed-control mechanism itself is independently verified sound, but effect sizes attributable to any single design choice have not been confirmed stable across additional seeds.
3. Pooled KS-statistic. The distributional-fidelity metric in Table 2 pools all 36 channels into one comparison and likely carries a volume-weighting bias analogous to the one already corrected for MAE; a channel-normalized macro-averaged KS-statistic is recommended future work, not yet implemented.
4. Ground-truth data quality. PhysioNet 2012 is documented to contain data-entry artifacts. This paper provides concrete, twice-reproduced evidence for one such artifact (arterial pH, Section 7) but has not systematically screened all 36 channels for similar issues.
5. Demographic bias not assessed. The stratified analysis in this paper examines mortality outcome only; disparity across demographic subgroups (age, gender, admission type) is not evaluated.
6. Downstream-utility method differences are not significance-tested. The overlapping standard deviations in Table 6 suggest the observed AUROC/AUPRC differences between methods may not be statistically distinguishable from fold-to-fold variance; this paper reports point estimates but does not perform a formal significance test between methods, and the results in Section 6.5 should not be over-interpreted as a method ranking.
7. Deferred scope, stated explicitly in Section 1.3.3: Mahalanobis-based multivariate plausibility checking, the TIMBA/Mamba architecture family, RMSE/MRE/Wasserstein distance, Cubic Spline interpolation,

and evaluation on the full 12,000-patient PhysioNet 2012 release.

10 Future Work

1. Re-execute the gradient-clipping ablation against the exact hyperparameter configuration used for the primary benchmark reported here, across at least two random seeds, before drawing any conclusion about clipping's effect under this configuration.
2. Systematically screen all 36 channels' ground-truth values in the test fold for implausible outliers, following the pattern identified for pH in Section 7.
3. Implement a channel-normalized, macro-averaged KS-statistic to close the same volume-weighting gap already addressed for MAE.
4. Evaluate BRITS and SAITS under block (multi-hour) missingness, with each channel's block placed independently so cross-channel information remains available during any single channel's gap, and with the validation fold's masking scheme updated to match — the appropriate next test of whether cross-channel correlation provides genuine value once local interpolation is structurally unavailable, and one that should follow, not precede, resolution of the items above.
5. Apply a formal significance test (e.g., a paired bootstrap over cross-validation folds) to the downstream-utility results in Table 6 before treating any pairwise method difference as a confirmed finding.
6. Address the remaining deferred scope (Mahalanobis plausibility, TIMBA/Mamba, RMSE/MRE/Wasserstein, full 12,000-patient evaluation) as a defined subsequent project stage.

11 Conclusion

Under an evaluation metric that treats all 36 PhysioNet 2012 clinical channels with equal importance, the literature-reported advantage of BRITS and SAITS over classical imputation baselines does not reproduce: both deep sequential imputers underperform even a constant-mean baseline, while deterministic methods lead decisively on both pointwise and distributional fidelity. Per-channel analysis attributes this to a combination of uneven training signal across channels, nonlinear error amplification in log-transformed heavy-tailed channels, and at least one confirmed ground-truth data-quality artifact independently reproduced across two experimental runs. A stratified extension finds a consistent, method-independent reconstruction disparity disadvantaging non-surviving patients — a finding with direct relevance to equitable clinical deployment.

A second extension, measuring downstream clinical utility, found that no method or method class shows a consistent advantage in downstream mortality-classification utility, and that a method's pointwise reconstruction accuracy does not reliably predict its downstream utility — a further illustration of this paper's central argument that different evaluation questions can yield materially different, and sometimes uncorrelated, answers. We regard the explicit reconciliation between planned and executed scope throughout this paper as inseparable from its technical contribution: in an evaluation-heavy field, the rigor of the evaluation pipeline is not incidental to the result, it frequently is the result.

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A Appendix: Physiological Bounds

Channel	Range	Channel	Range
ALP	0–2000	ALT	0–5000
AST	0–5000	Albumin	0.5–6.0
BUN	0–300	Bilirubin	0–50
Cholesterol	0–500	Creatinine	0–20
DiasABP	0–200	FiO2	0.2–1.0
GCS	3–15	Glucose	0–1000
HCO3	0–60	HCT	0–80
HR	0–300	K	1.0–10.0
Lactate	0–30	Mg	0.5–10.0
MAP	0–250	MechVent	0–1
Na	100–200	NIDiasABP	0–200
NIMAP	0–250	NISysABP	0–300
PaCO2	0–150	PaO2	0–700
pH	6.5–8.0	Platelets	0–1000
RespRate	0–80	SaO2	0–100
SysABP	0–300	Temp	30–45
TroponinI	0–50	TroponinT	0–10
Urine	0–2000	WBC	0–200

Table 7: Physiological bounds used for output clamping. These are wide engineering guardrails against implausible model outputs, not clinical reference ranges, and have not been reviewed by a clinician.